PLAN-08 - Testing, Verification, and Acceptance

POS-DGFY - Global Location Switching | Plan v1.0 | Revision 2026-09-27

Field	Value
Project	POS-DGFY - Global Location Switching
Plan version	1.0
Section	PLAN-08 - Testing, Verification, and Acceptance
Revision date	2026-09-27
Plan status	NOT EXECUTION READY
Purpose	Provide the verification matrix and concrete tests required before the plan can be marked execution complete.
Dependencies	PLAN-04 through PLAN-07; repository test conventions; web-performance-qa and relevant POS E2E skills.

Verification versus validation

Verification proves the implementation follows the approved requirements. Validation proves that users can safely operate multiple branches without confusion or cross-location errors.

Verification matrix

ID	Requirement or rule	Method	Evidence
TEST-001	FR-001 / FR-002	Desktop and mobile UI	Screenshots/video showing explicit operating/viewing labels.
TEST-002	FR-003 / BR-004	Backend integration	Atomic close/open transition and persisted audit event.
TEST-003	FR-004 / SEC-001 / SEC-002	Authorization API	403 for unassigned or forged destination.
TEST-004	BR-003	Failure-path integration	Pending offline/cart/payment blocks switch without mutation.
TEST-005	FR-005 / DATA-001 / DATA-002	Database and E2E	Sale, stock movement, queue, and report all use one location.
TEST-006	FR-006 / BR-007	Report regression	Historical location remains unchanged after switch.
TEST-007	FR-007	Module contract tests	Settings, voucher, pricing, affiliate scope is enforced.
TEST-008	NFR-001 / NFR-002	Concurrency test	No stale overwrite and duplicate transition is idempotent.
TEST-009	MIG-001	Migration rehearsal	Counts, constraints, ownership, reports, and rollback pass.
TEST-010	OPS-001	Observability test	Metrics, logs, alerts, and audit evidence appear.

Required GIVEN / WHEN / THEN scenarios

- Given a user has access to Location A and B, when they switch with a clear cart and valid cash values, then the new shift and terminal context are Location B.
- Given a user lacks Location B access, when they submit destination B directly to the API, then the server returns 403 and Location A remains unchanged.
- Given pending offline operations, when the user selects another location, then the switch is blocked and no shift or cash state changes.
- Given two terminals target the same destination shift, when both submit, then one succeeds and the other receives a deterministic conflict.
- Given a sale completed at Location A, when an administrator views Location B, then the sale remains attributed to A.

- Given a parked sale belongs to Location A, when a cashier works at Location B, then it cannot be silently paid as a B sale.

Test layers

Layer	Coverage
Unit	Scope policy, context reducer, transition validation, formatting, permission decisions.
API	Authorization, validation, idempotency, conflicts, tenant and location ownership.
Database	Constraints, locks, historical snapshots, inventory movements, migration parity.
UI	Selector, modal, disabled states, labels, mobile/desktop responsive behavior.
E2E	Login -> context selection -> shift -> sale -> stock -> report -> audit -> close.
Security	Direct API bypass, parameter tampering, cross-tenant and cross-location attempts.
Recovery	Timeout, retry, refresh, restart, failed migration, rollback, reconciliation.
Performance	Large item lists, reports, location filters, concurrent users, refresh storms.

Acceptance gate

No critical requirement may be marked verified without implementation evidence and verification evidence. A passing component test does not prove cross-module correctness.